

Human3D Segmentation

Installation Steps:

1. Clone the github repo:

2. Your system must have docker, docker compose and nvidia container toolkit installed
You can install them from these guides

- <https://docs.docker.com/engine/install/ubuntu/>
- <https://docs.docker.com/compose/install/linux/>
- <https://docs.nvidia.com/datacenter/cloud-native/container-toolkit/latest/install-guide.html>

3. Go to the docker folder in the code directory and build the docker image (it will take some time for the first time only) by typing in the terminal

- Sudo docker compose up -d

4. After finishing successfully go inside the container by typing

- Sudo docker exec -it 'container name' /bin/bash
- Replace the container name with your container name

5. Inside the docker container install the following

- cd /app/third_party/MinkowskiEngine
- python setup.py install --force_cuda --blas=openblas
- cd /app/third_party/pointnet2
- python setup.py install
- cd /app
- pip install pytorch-lightning==1.7.2
- pip3 install torch==1.12.1+cu113 torchvision==0.13.1+cu113 --extra-index-url <https://download.pytorch.org/whl/cu113>
- cd /app/docker
- pip install -r requirements.txt
- Pip install fvcore
- Pip install open3d==0.16.0

6. Download the models checkpoints by running:

- bash ./download_checkpoints.sh

7. Now you can run the script that predict your ply files and don't forget to make the log file mentioned in our meeting

Some Notes:

- Building the image is done once in the system
- If you removed the docker container you will need to repeat the installations inside it again
- This command removes the docker container
“docker compose down”